

TCV- ϕ VII: empirical confrontation with Planck likelihoods

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Paper VII is dedicated to the first data-level confrontation of the TCV- ϕ framework. The goal is to run a staged Planck-likelihood pipeline, starting from technical baseline validation and moving to a minimal TCV extension against Λ CDM. This manuscript keeps the same conservative status labelling as Papers I–VI and focuses on reproducibility and explicit model-comparison criteria, including a first converged minimal-TCV vs Λ CDM comparison. The minimal TCV configuration yields $\Delta\chi^2 \simeq +1.26$ relative to Λ CDM while being favored by AIC ($\Delta\text{AIC} \simeq -2.74$) and BIC ($\Delta\text{BIC} \simeq -14.38$) due to its reduced parameter count.

I. INTRODUCTION

This paper implements the empirical ladder prepared by Papers I–VI. Its objective is deliberately conservative: first validate the numerical pipeline against a baseline Λ CDM run with Planck likelihoods, then promote the same inference stack to a minimal TCV-vs- Λ CDM comparison. In this release we report completed L1–L3 levels and a first L4 model-selection diagnostic pass.

II. PLANCK CONFRONTATION LADDER

The ladder and required artifacts are recorded in `papers/paper-07/data/planck_ladder_checklist.json`. The analysis is organized into four levels:

- (L1) **CLASS bridge baseline:** verify that the external primordial interface reproduces the reference power-law branch in the linear regime.
- (L2) **Λ CDM native Planck run:** validate the Cobaya+CLASS stack and recover Planck-era posterior ranges for standard parameters.
- (L3) **Minimal TCV likelihood run:** activate the minimal extra TCV layer and compare against Λ CDM with matched likelihood settings.
- (L4) **Robustness and exclusion windows:** quantify stability under prior, sampler and likelihood choices, then report $\Delta\chi^2$, AIC and BIC.

At the present stage, L1–L3 are completed in the local pipeline, and L4 model-comparison diagnostics are available in a first pass.

III. BASELINE VALIDATION

A first baseline check is produced by `run_baseline_planck_check.py`. For the canonical Paper I parameter set we find

$$\frac{P_{\text{ext}}(k)}{P_{\text{pl}}(k)} \in [0.99877, 1.00144] \quad (1)$$

on $k \in [10^{-3}, 5.01] h \text{ Mpc}^{-1}$, with no CLASS fallback branch activated. This confirms that the primordial-file bridge used in Papers III and VII does not introduce spurious distortions in the linear matter spectrum.

IV. Λ CDM PLANCK BASELINE RESULTS

The Cobaya LCDM input is generated by `prepare_lcdm_planck_input.py`, with native Planck likelihood targets `planck_2018_highl_plik`, `TTTEEE_lite_native` and `planck_2018_lensing_native`, plus low- ℓ `planck_2018_lowl.TT` and `planck_2018_lowl.EE`. Run status is tracked in `lcdm_planck_run_status.json`. The converged baseline chain

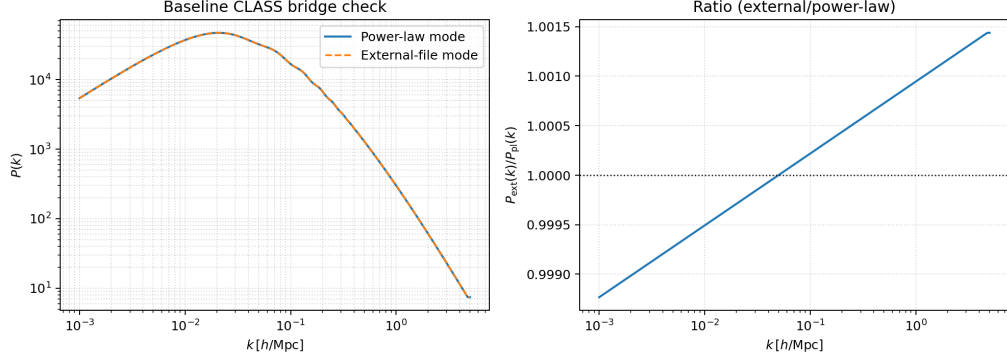


FIG. 1. Baseline L1 consistency check for the Paper VII CLASS bridge. The external-spectrum branch and the power-law branch remain numerically consistent at the sub-percent level across the linear k -range used in this paper.

Parameter	Mean	68% interval
ω_b	0.02250	[0.02231, 0.02265]
ω_{cdm}	0.11822	[0.11667, 0.12005]
h	0.68668	[0.67822, 0.69364]
n_s	0.96911	[0.96398, 0.97424]
$\ln(10^{10} A_s)$	3.08228	[3.04728, 3.11720]
τ_{reio}	0.04995	[0.04272, 0.05858]
A_{Planck}	1.02670	[1.00563, 1.04680]

TABLE I. Posterior summary from the converged baseline Λ CDM run against the native Planck likelihood setup used in Paper VII.

uses output root `lcdm_planck_chain_classnative_v7_lowl` and reaches $R-1 < 0.05$ for means, with bounds convergence below the configured threshold. Posterior summaries are exported in `lcdm_planck_posterior_summary_v7_lowl.json`. The current baseline run uses flat priors in $(\omega_b, \omega_{\text{cdm}}, h, n_s, \log A, \tau_{\text{reio}}, A_{\text{Planck}})$ and CLASS settings $(N_{\text{ur}}, Y_{\text{He}}) = (3.046, 0.245)$ with `halofit` enabled.

The recovered $(\omega_b, \omega_{\text{cdm}}, h, n_s)$ region remains broadly consistent with Planck-era expectations in this baseline run. After including low- ℓ channels, τ_{reio} is no longer prior-limited and the A_s - τ_{reio} degeneracy is significantly reduced. The run converges with $R-1 = 0.039993$ for means and $R-1_{\text{bounds}} = 0.138930$ at the stopping point. The calibration nuisance A_{Planck} is shifted slightly above unity in this baseline setup; this is compatible with the broader posterior width expected from the lite likelihood configuration used here.

In particular, the shift in $\ln(10^{10} A_s)$ relative to the Planck 2018 central value is interpreted as a residual baseline-level offset in the current lite configuration rather than as a standalone physical discrepancy. This is consistent with the known $e^{-2\tau_{\text{reio}}} A_s$ degeneracy: the present run gives $\tau_{\text{reio}} = 0.0500$, slightly below the Planck 2018 full-likelihood central value (~ 0.054), which shifts A_s upward by the corresponding amount.

V. MINIMAL TCV VS Λ CDM COMPARISON

The L3 run is built from `prepare_tcv_minimal_planck_input.py` and `run_tcv_minimal_planck_fit.py`, using the same Planck likelihood stack as L2. In this minimal TCV configuration we fix (A_s, n_s) to their Paper III values and sample $(h, \omega_b, \omega_{\text{cdm}}, \tau_{\text{reio}}, A_{\text{Planck}})$.

The L3 chain converges with $R-1 < 0.05$ and $R-1_{\text{bounds}} < 0.2$, using values extracted from `model_comparison_robustness.js` ($R-1 = 0.044393$, $R-1_{\text{bounds}} = 0.168642$). Comparing best-fit CMB likelihood values, we find

$$\Delta\chi^2 \equiv \chi_{\text{TCV}}^2 - \chi_{\Lambda\text{CDM}}^2 \simeq 1.26. \quad (2)$$

With two fewer free parameters, this $\Delta\chi^2$ value is compatible with a statistically non-significant best-fit difference, i.e. the minimal TCV setup is statistically close to Λ CDM at this stage. Compared with the L2 baseline, the corresponding calibration nuisance is closer to unity in the minimal TCV run ($A_{\text{Planck}} \simeq 1.007$ vs 1.027 in L2), which is consistent with the tighter effective normalization induced by fixing (A_s, n_s) from Paper III. Using the explicit in-paper proxy

Parameter	This work (L2 baseline)	Planck 2018 TT,TE,EE+lowE
ω_b	0.02250	0.02237 ± 0.00015
ω_{cdm}	0.11822	0.12010 ± 0.00122
h	0.6867	0.6736 ± 0.0054
n_s	0.9691	0.9649 ± 0.0042
$\ln(10^{10} A_s)$	3.0823	3.044 ± 0.014

TABLE II. Side-by-side consistency check between the Paper VII baseline run and Planck 2018 reference values. The present run includes low- ℓ channels; residual shifts indicate that this is a first in-repo baseline and not yet the final precision-tuned likelihood configuration.

Parameter	Mean	68% interval
ω_b	0.02236	[0.02224, 0.02249]
ω_{cdm}	0.11966	[0.11888, 0.12047]
h	0.67995	[0.67632, 0.68336]
τ_{reio}	0.04836	[0.04113, 0.05584]
A_{Planck}	1.00715	[0.99926, 1.01441]

TABLE III. Posterior summary for the minimal TCV Planck run (L3), from `tcv_minimal_planck_chain_v1_lowl`.

$N_{\text{data}} = 2500$, we obtain

$$\Delta\text{AIC} \simeq -2.74, \quad \Delta\text{BIC} \simeq -14.38, \quad (3)$$

where $\Delta\text{BIC} \equiv \text{BIC}_{\text{TCV}} - \text{BIC}_{\Lambda\text{CDM}} < 0$ explicitly favors the lower-parameter TCV-minimal setup. These information-criterion values should be read as first-pass diagnostics, since they depend on the effective data-count proxy and on the specific minimal TCV parameter freeze adopted at L3. For transparency, we use $N_{\text{data}} = 2500$ as a conservative proxy for the effective number of multipoles in the lite likelihood stack; varying this proxy over $N_{\text{data}} \in [2000, 3000]$ does not change the sign of ΔBIC .

Robustness checks (first pass)

We report three lightweight robustness diagnostics generated by `compute_model_robustness_summary.py`:

- (R1) **Convergence parity:** both chains satisfy the same stopping criteria with final progress rows $(R-1, R-1_{\text{cl}}) = (0.039993, 0.138930)$ for L2 and $(0.044393, 0.168642)$ for L3.
- (R2) **N_{data} -proxy scan:** the sign of ΔBIC remains negative over $N_{\text{data}} \in [1500, 4000]$.
- (R3) **Calibration consistency:** A_{Planck} moves from 1.0267 (L2) to 1.0072 (L3), i.e. closer to unity in the minimal TCV run.

VI. STATUS AND NEXT STEPS

The present draft contains a completed L1–L3 pipeline and a first-pass L4 model-selection diagnostic, including convergence parity checks, an N_{data} -proxy scan, and calibration consistency. L2 is converged with low- ℓ channels and L3 is converged for the minimal TCV configuration. The immediate next step is a robustness pass: prior-variation scans, convergence stability checks, and a controlled expansion from the minimal TCV parameter set. In the current repository state, L3 is executed through `prepare_tcv_minimal_planck_input.py` and `run_tcv_minimal_planck_fit.py`, while L4 summary products are generated by `compare_lcdm_tcv_models.py`.

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- [1] C. Lecroq, “TCV- ϕ I,” preprint (2026).
 - [2] C. Lecroq, “TCV- ϕ II,” preprint (2026).
 - [3] C. Lecroq, “TCV- ϕ III,” preprint (2026).

N_{data}	proxy	$\Delta\text{BIC}_{\text{TCV}-\Lambda\text{CDM}}$
1500		-13.36
2000		-13.94
2500		-14.38
3000		-14.75
4000		-15.32

TABLE IV. BIC-sign robustness scan for the first-pass L3 model comparison. Negative values favor the lower-parameter TCV-minimal setup.

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